

POLYP DETECTION MEDICAL REPORT

Analysis Date: June 04, 2026

VIDEO INFORMATION

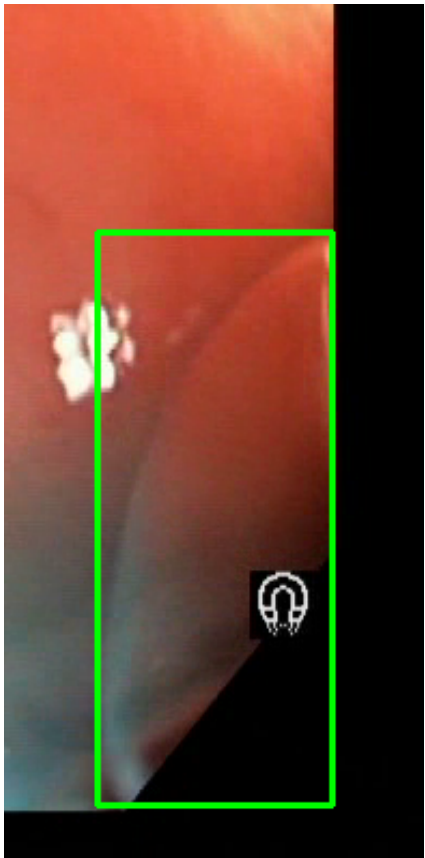
Video ID:	7cfdbb45-a132-4a04-8e6e-72270e3c7792
Total Frames:	7784
Frame Rate:	25.0 fps (assumed)
Duration:	311.4 seconds

DETECTION SUMMARY

Detection Method	Count	Status
YOLO Detections	6380	✓
RT-DETR Detections	6997	✓
MedSAM2 Detections	6923	✓
Consensus (3-Model Agreement)	3	✓
Risk Classification	Count	Percentage
HIGH RISK	0	0.0%
MEDIUM RISK	3	100.0%
LOW RISK	0	0.0%
TOTAL ANALYZED	3	100.0%

DETAILED POLYP ANALYSIS

POLYP #1 — SERRATED_POLYP [MEDIUM_RISK] (Tracked across 175 frames)



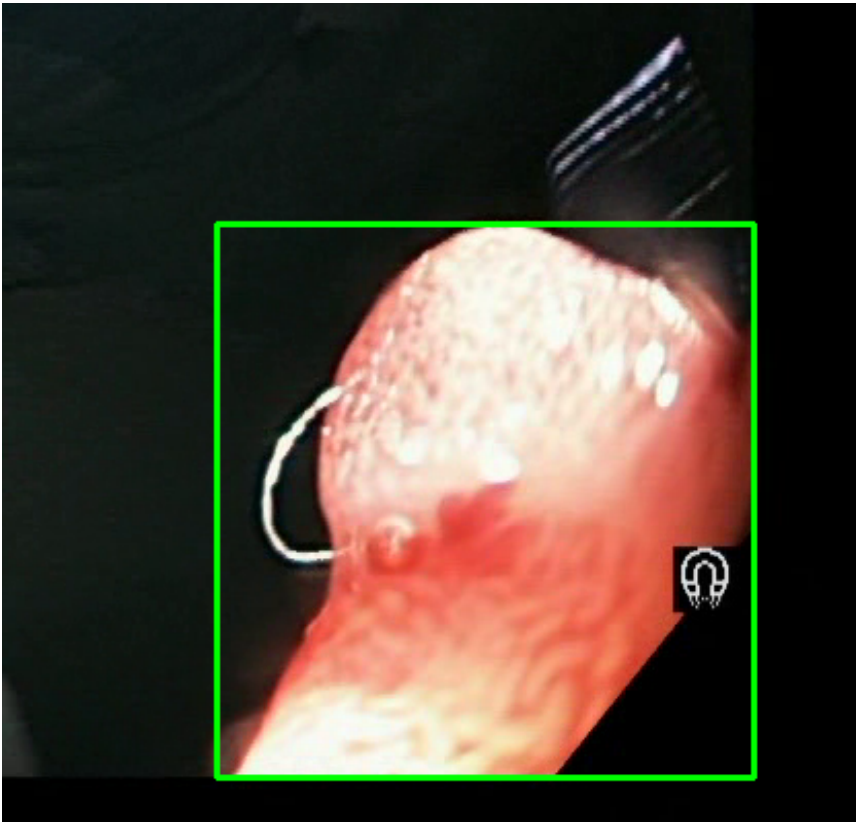
Feature	Value	Interpretation
Frame Index	4315	Frame 4315 of 7784
Detection Method	Detector Consensus (MedSAM2 Unavailable)	Detector Consensus (MedSAM2 Unavailable) (MedSAM2 Unavailable for this video)
POLYP TYPE	SERRATED_POLYP	Type confidence: 92.0%
Redness Score	0.126	Color-based vascularization
Relative Radius	39.3%	Polyp size as % of frame diagonal
Texture Score	0.234	Surface morphology
Vessel Visibility	0.828	Vascularization indicator
Risk Tier	MEDIUM RISK	Validated by ESGE 2017 / NICE classification

Clinical Classification	LOW_RISK	Final symbolic reasoning bucket
Expert Prediction	LOW_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	90.0%	YOLO / RT-DETR / track confidence

Type Assessment: Serrated lesion — consider serrated polyposis syndrome. Complete resection required.

Low-confidence MODERATE RISK detection - Clinical review recommended

POLYP #2 — FLAT_POLYP [MEDIUM_RISK] (Tracked across 5662 frames)



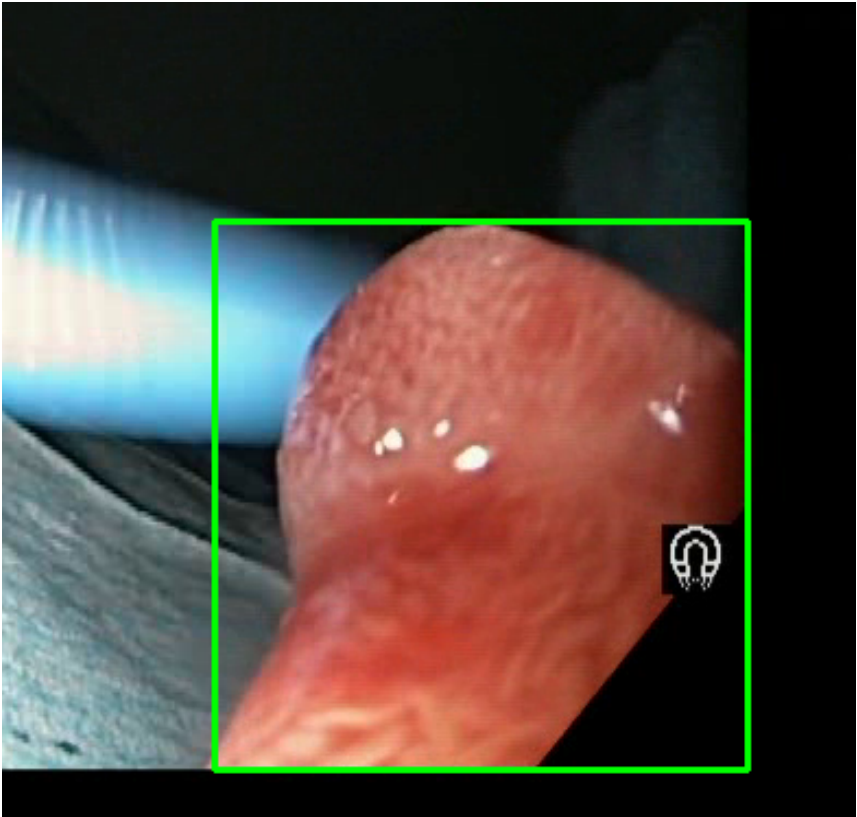
Feature	Value	Interpretation
Frame Index	7506	Frame 7506 of 7784
Detection Method	Detector Consensus (YOLOv8 / RT-DETR)	Model Agreement (MedSAM2 unavailable for this video)
POLYP TYPE	FLAT_POLYP	Type confidence: 92.0%
Redness Score	0.089	Color-based vascularization

Relative Radius	16.3%	Polyp size as % of frame diagonal
Texture Score	0.183	Surface morphology
Vessel Visibility	0.873	Vascularization indicator
Risk Tier	MEDIUM RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	91.6%	YOLO / RT-DETR / track confidence

Type Assessment: Flat mucosal lesion — chromoendoscopy or NBI assessment recommended.

Low-confidence MODERATE RISK detection - Clinical review recommended

POLYP #3 — FLAT_POLYP [MEDIUM_RISK] (Tracked across 197 frames)



Feature	Value	Interpretation
Frame Index	7730	Frame 7730 of 7784

Detection Method	Detector Consensus (YOLO / RT-DETR)	MedSAM2 Unavailable (MedSAM2 unavailable for this video)
POLYP TYPE	FLAT_POLYP	Type confidence: 84.8%
Redness Score	0.170	Color-based vascularization
Relative Radius	35.1%	Polyp size as % of frame diagonal
Texture Score	0.243	Surface morphology
Vessel Visibility	0.117	Vascularization indicator
Risk Tier	MEDIUM RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	NORMAL_MUC OSA	Normal mucosal pattern — no polyp morphology identified
Detection Confidence	91.8%	YOLO / RT-DETR / track confidence

Type Assessment: Flat mucosal lesion — chromoendoscopy or NBI assessment recommended.

Low-confidence MODERATE RISK detection - Clinical review recommended

CLINICAL IMPRESSION

Summary:

A total of 3 polyps were detected and analyzed using multi-model consensus and AI-based risk stratification.

Risk Distribution:

- HIGH RISK polyps: 0 (0.0%)
- MEDIUM RISK polyps: 3 (100.0%)
- LOW RISK polyps: 0 (0.0%)

Recommendation:

The presence of MEDIUM RISK polyps warrants clinical review and correlation with endoscopic findings.

Technical Details:

- Detection: Multi-model consensus (YOLO, RT-DETR, MedSAM2)
- Features: 444-dimensional vectors (384 SSL + 60 biomarkers)
- Classification: Cluster-specific Decision Tree experts
- Confidence: Based on deep learning + medical heuristics

Disclaimer: This report is generated by an AI-assisted research system for academic and research purposes only. All clinical recommendations are derived from published ESGE 2017, NICE CG118/CG131, and BSG 2019 colonoscopy guidelines. This output does not constitute medical advice and must not be used as the sole basis for clinical decisions. All findings must be reviewed and confirmed by a qualified gastroenterologist or endoscopist.